# Question 1

Create a program for a computer shop, accept the computer component being bought, price of the component and the quantity. The program must calculate the total cost and display whether the purchase is a Bulk purchase (R2000 or more) or a normal purchase (less than R2000).

1.1. Create an IPO chart using the above scenario (4)

1.2. Use meaningful variable names, correct data types, formulas and formatted output to prompt.

1.2.1. The component name. (1)

1.2.2. The unit price in rand. (1)

1.2.3. The quantity purchased. (1)

1.2.4. Convert the price to float and quantity to int. (1)

1.2.5. Remove unnecessary spaces from the component name and use meaningful variable names. (1)

1.2.6. Calculate the total cost using the correct formula. (1)

1.2.7. Store the result in a variable. (1)

1.2.8. Use a conditional statement to determine the purchase category. (1)

1.2.9. Use the correct comparison operator (1)

1.2.10. Display a neatly formatted summary containing: (4)

- Component name.
- Unit price with two decimal places.
- Quantity.
- Total cost in rand with two decimal places.
- Purchase category.

1.3. Include a header comment containing the learner's name, date and purpose of the program. (1)

1.4. Use the following test data and record the expected and actual outputs. (2)

| Component | Price | Quantity | Expected total | Category | Expected outputs | Actual outputs |
|---|---|---|---|---|---|---|
| **Keyboard** | R250.00 | 2 | R500.00 | Normal Purchase | | |
| **RAM** | R750.00 | 3 | R2,250.00 | Bulk Purchase | | |
| **SSD** | R1,200.00 | 2 | R2,400.00 | Bulk Purchase | | |

## Question 2

2.1. What would happen if the programmer did not convert the unit price input to a float before multiplying it by the quantity
(2)

2.2. Consider the expression below:
total_cost = unit_price + quantity * 2
If unit_price = 100 and quantity = 3,
what is the result, and why?